Sarah Yuzi Sandström

Computer Science student · Python · SQL · Data pipelines · Testing & CI

Kauniainen, Finland · Permanent Finnish residency · +358 45 102 9420
sarahyuzisandstrom@gmail.com · LinkedIn · GitHub · https://yuzi2021.github.io/portfolio-data-ai/

PROFILE

Computer Science student (University of London, expected 2027) who has been building software through my studies and independent projects since 2023, after an established career in education. My recent work covers Python, SQL and PostgreSQL, relational data modelling, REST APIs, validated ETL, backend permissions and WebSockets, automated testing, GitHub Actions, Docker and deployment. I am working towards data and AI engineering and want to develop this foundation in a professional engineering environment.

TECHNICAL SKILLS

Programming & data	Python · SQL · PostgreSQL · SQLite · relational data modelling · data validation & transformation
Backend & applications	Django · DRF · REST APIs · auth & permissions · Channels/WebSockets · Daphne/ASGI · Redis/Celery (local Docker) · JavaScript · HTML/CSS
Development & quality	Git · GitHub · GitHub Actions · CI/CD · automated testing (Django, unittest, pytest) · Docker/Compose · Render · health checks
Data & AI	ETL · scheduled ingestion & change detection · TensorFlow/Keras · scikit-learn · SHAP · NumPy · pandas

PROJECTS

SkillPath Navigator — data pipeline & REST API

Coursework, extended independently · 2026

Live demo · source repository private

- Django REST Framework API over 464 job postings, 319 skills and 5,845 job–skill relationships, with a many-to-many model through an explicit join table.
- Built an extract → validate → transform → load command: malformed rows, conflicting duplicates and broken cross-file references reject the snapshot before an atomic, repeatable load.
- 44 passing tests, run by GitHub Actions against PostgreSQL 16; Docker Compose for development; SQLite demo deployed on Render only after checks pass.

Phishing email classifier: signal or artifact?

CM3015 Machine Learning coursework · 2026

- Parsed the Nazario (mbox) and Enron (maildir) corpora into one record format; deduplicated before splitting, balanced and stratified the data; fitted the vocabulary on training data only.
- TensorFlow Dense/Dropout models compared across three seeds; test F1 0.970–0.998. SHAP showed 18 of the 25 most influential tokens in the headers model were source artifacts, and recall fell to 0.38 on other benchmark corpora.

eLearning Platform — full-stack & backend engineering

University project, productionised independently · 2026

Live demo · case study · source repository private

- Redesigned a role-aware Django/DRF learning platform with authentication and ownership permissions across HTML, REST, protected resources and Channels WebSockets.
- Deployed with Daphne/ASGI and persistent Neon PostgreSQL; added health, liveness and readiness endpoints.
- 85/85 tests passing in GitHub Actions CI. PostgreSQL, Redis, Celery and distributed Channels are implemented/tested in the full local Docker stack; Redis/Celery are not deployed publicly.

Other technical work: GRC Change Monitor (six-source GitHub Actions ingestion, 67 deterministic tests) · SMS spam classification with Naïve Bayes implemented from scratch in NumPy · AI research digest agent · GRC Ops Lab (CSV reconciliation, 35 pytest tests)

PROFESSIONAL EXPERIENCE

Privacy & Digital-Workplace Development Project

2024–2026

- Investigated digital-tool and information-handling problems with staff, management, IT and a Data Protection Officer; produced a legally reviewed 57-page staff guide and training. Published as a co-authored thesis (Laurea, 2024).

Teacher Assistant — Pilke Päiväkodit, Finland

2022–2024

- Resolved issues independently in a fast-moving service environment; designed and delivered staff learning on data protection, information security and responsible use of digital tools.

Educator / Pedagogy Consultant — International schools & education technology

2013–2022

- At LingoAce, automated the reporting system, reducing reporting time by 70%. Developed learning materials and worked across multicultural teams, adapting explanations to each audience.

EDUCATION

BSc Computer Science — University of London

Expected 2027

Web development · databases · APIs · software testing · machine learning

Master's studies in Changing Education, University of Helsinki (2025) · MA English Studies & Subject Teacher Training, University of Helsinki (2019) · BA English Studies for Teaching, University of Buckingham, First Class Honours (2013)

Languages: Mandarin (native) · English (full professional) · Finnish (basic to pre-intermediate) · Swedish (elementary)